

CivicPulse AI

AI-Powered Emergency Incident Response & Dispatch Orchestration

UiPath AgentHack. Built the AI agents of tomorrow.



AI-Powered Emergency Incident Response & Dispatch Orchestration



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Problem statement and proposed solution

Problem

- Emergency incidents are often handled through manual coordination, resulting in delayed response, inconsistent prioritization, and fragmented communication between departments.
- Critical decisions such as risk assessment, SLA assignment, dispatch routing, and incident closure rely heavily on human intervention, increasing response time and the likelihood of operational errors.
- Manual documentation and closure reporting create additional administrative overhead and reduce audit readiness.

Solution

- **CivicPulse AI** is an AI-powered emergency response orchestration platform built on **UiPath Maestro** that automates the complete incident lifecycle—from incident review to AI-generated closure reports.
- AI Agents perform **risk assessment, SLA assignment, priority classification, and incident closure analysis**, while Maestro orchestrates approvals, dispatch workflows, and human-in-the-loop decisions.
- The solution integrates **UiPath Apps, UiPath agents, Data Fabric, RPA, email notifications, and structured reporting** to deliver faster response times, improved operational visibility, and standardized incident management.

Value Proposition:

Accelerating emergency response through AI-driven decision making, intelligent orchestration, and end-to-end automation.

Benefits and technologies used

Details

End-user	Emergency Control Room Operators, Dispatch Teams, Incident Response Coordinators
User department	Public Safety, Emergency Services, Smart City Operations, Disaster Management Centers
Industries	Government, Smart Cities, Public Safety, Healthcare, Transportation, Critical Infrastructure
UiPath products used	UiPath Maestro, AI Agents, Action Center, Data Fabric, Integration Service, Orchestrator, RPA Workflows
Other - integrations / APIs / technologies used	Gmail, Storage Buckets, BPMN Workflow, AI-based Risk & Incident Analysis

Benefits, impact and outcomes

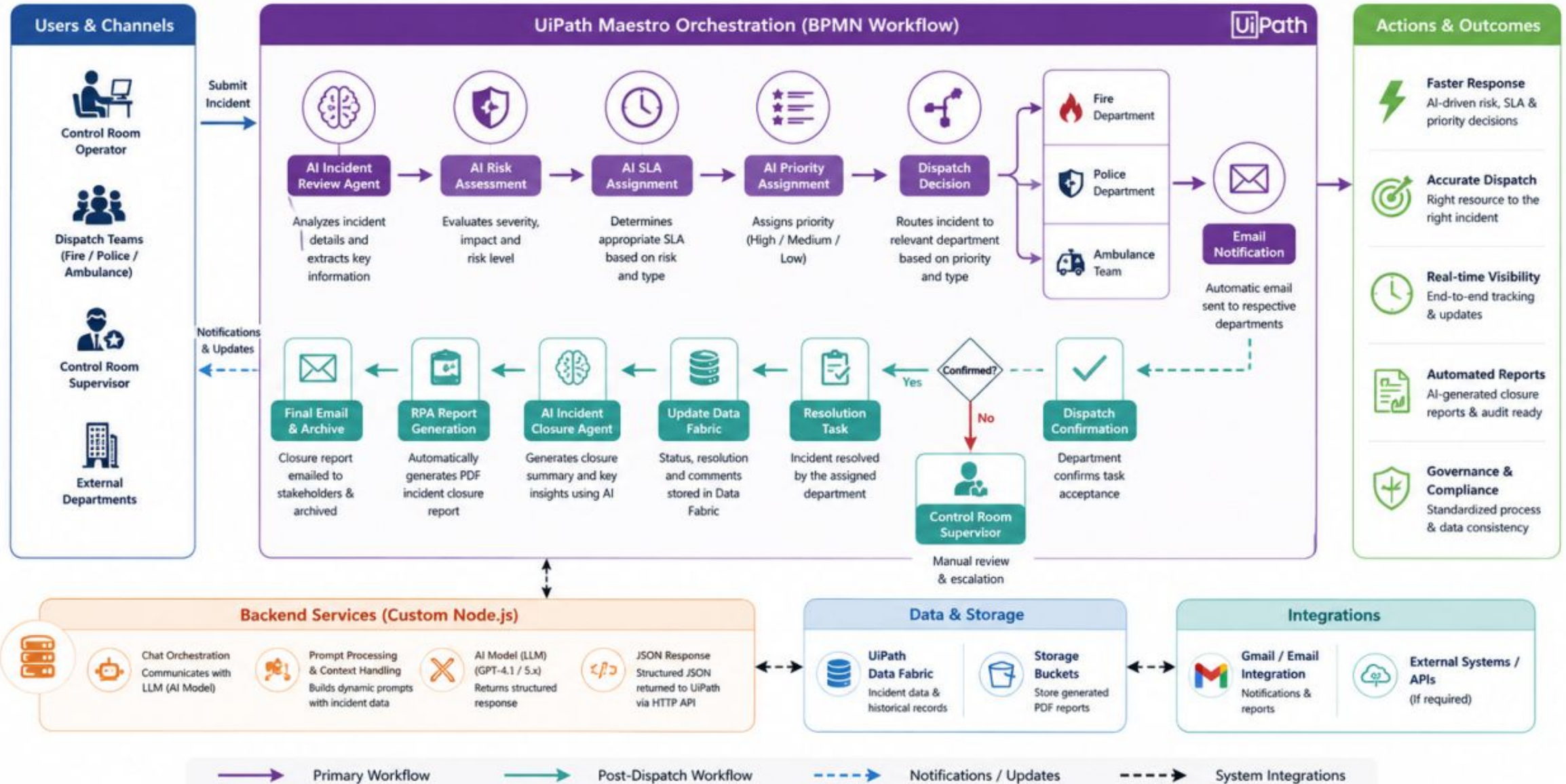
- ❑ **Automates the complete emergency incident lifecycle** from AI-powered incident review to closure reporting.
- ❑ **Combines AI orchestration with a custom Node.js backend**, enabling structured JSON responses through REST APIs for seamless integration with UiPath workflows.
- ❑ **Reduces manual intervention** by automating risk assessment, dispatch routing, notifications, and incident documentation.
- ❑ **Provides centralized, audit-ready records** using Data Fabric and automated report generation.
- ❑ **Delivers a scalable architecture** where additional AI services and external systems can be integrated through REST APIs.

Key Value Delivered:

✓ Faster Response ✓ AI-Assisted Decision Making ✓ End-to-End Automation ✓ Better Operational Visibility ✓ Audit-Ready Incident Management

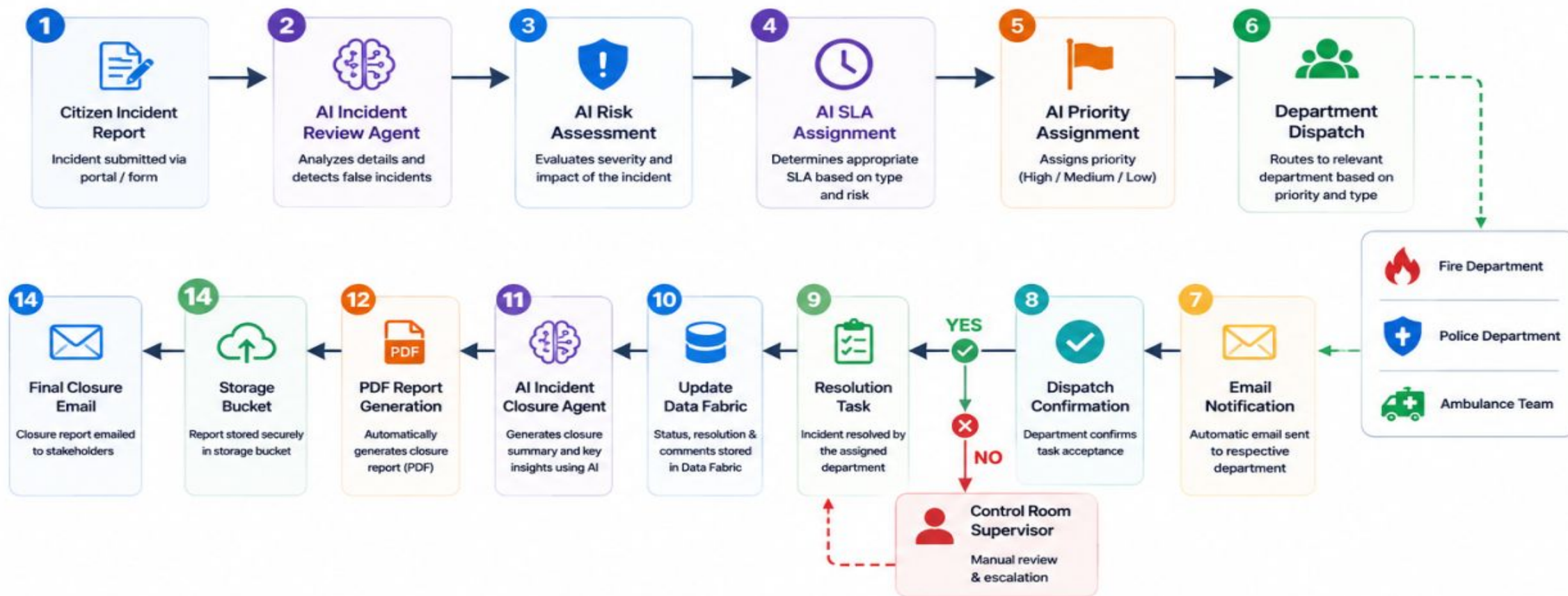
CivicPulse AI – Solution Architecture

AI-Powered Emergency Incident Response & Dispatch Orchestration



End-to-End Incident Lifecycle

AI-Powered • Orchestrated • Human-in-the-Loop



Faster Response

Reduced response time with AI-driven decisions



Accurate Dispatch

Right resource to the right incident



End-to-End Visibility

Real-time tracking and updates



Automated Reports

AI-generated reports for audit & compliance



Better Governance

Standardized process with human oversight



Fully orchestrated by UiPath Maestro with AI-assisted decision making and Human-in-the-Loop approvals.

AI Agents & Responsibilities

AI Component	Responsibility
AI Incident Review Agent	Reviews incident details and detects false incidents
AI Risk Assessment Agent	Calculates incident severity and risk
AI SLA Assignment Agent	Determines SLA based on incident type
AI Priority Agent	Assigns operational priority
AI Incident Closure Agent	Generates executive closure summary and audit-ready report
UiPath Maestro	Coordinates the complete BPMN workflow
Human Operators	Review dispatch confirmations and escalations

Human decisions remain in control while AI accelerates every operational step.

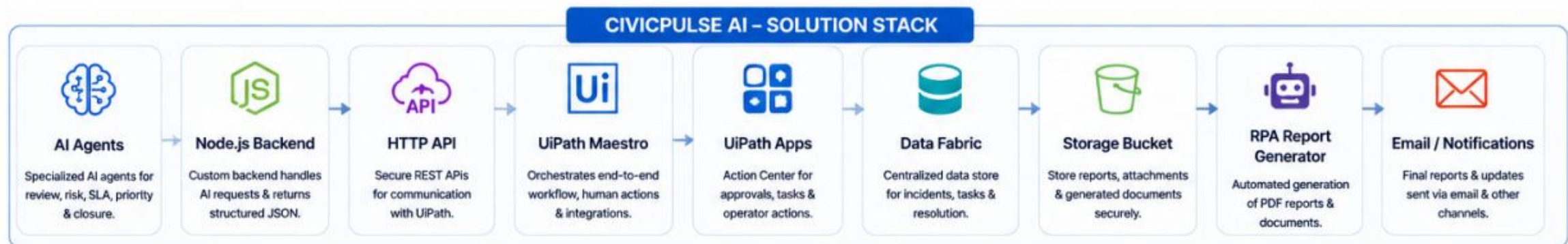
Future Roadmap

AI Improvements

- ❑ Predict incident escalation before it occurs
- ❑ AI-based department recommendation using historical incidents
- ❑ Intelligent resource optimization
- ❑ Dynamic ETA prediction
- ❑ AI-powered incident trend analytics dashboard

Enterprise Features

- ❑ Live GIS / Google Maps integration
- ❑ SMS & WhatsApp alerts
- ❑ IoT sensor integration
- ❑ Citizen mobile application
- ❑ Voice-based emergency reporting
- ❑ Power BI executive dashboard
- ❑ Real-time analytics
- ❑ Multi-language support



Our Vision: Smarter Predictions • Faster Response • Greater Safety • Connected Communities

THANK YOU.

CivicPulse AI- Smarter Response. Safer Communities

CivicPulse AI transforms emergency incident management with AI-powered intelligence, automation, and human collaboration- enabling faster decisions, seamless operations, and better outcomes when every second counts.



Faster Response

AI-driven decisions
reduce response time



Greater Safety

Right resources to
the right incident



End-to-End Automation

Seamless orchestration
from start to closure



Operational Visibility

Real-time tracking
and insights



Audit-Ready

AI-generated reports
for accountability

Together, let's build a future where technology and compassion work hand-in-hand
to protect lives and strengthen communities.